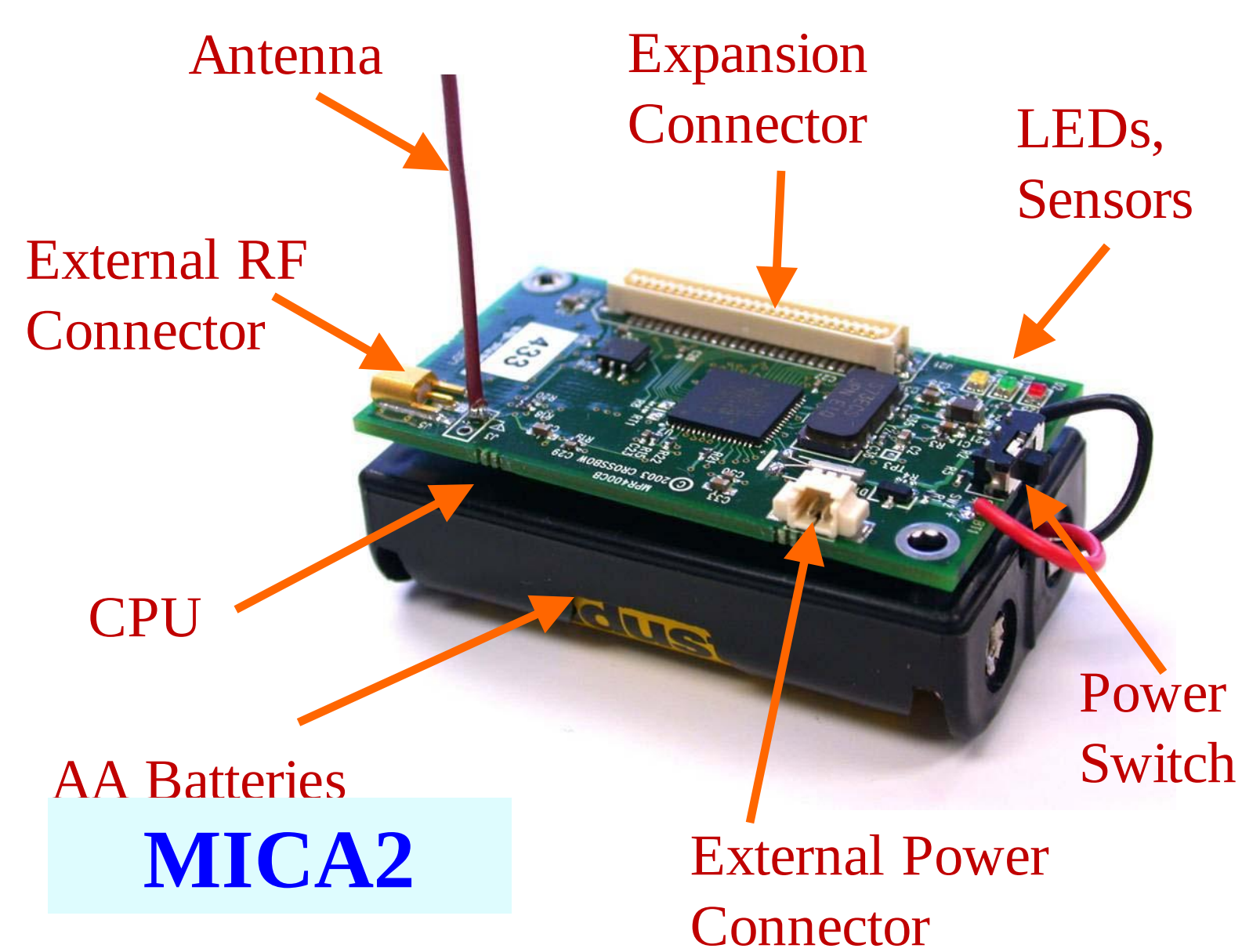


Wireless Sensor Networks

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Sensors with in-built Computers and Radio - MOTES



MICA2

□ Limited Energy

- Run on AA batteries

□ On-Board Sensors

- Humidity, Temperature, Light, Pressure, Accelerometer

□ Tiny Embedded Devices

□ Constrained Processors

- MPR 400 Atmel, TI MSP 430
- 4–8 MHz, 8-bit

□ Little Memory

- 128 KB Flash, 4–10 KB RAM

□ Multi-Channel Radios

- 900 MHz Chipcon CC1000
- 2.4 GHz Chipcon CC2420

□ Low Data Rate

- 40–250 Kbits/second

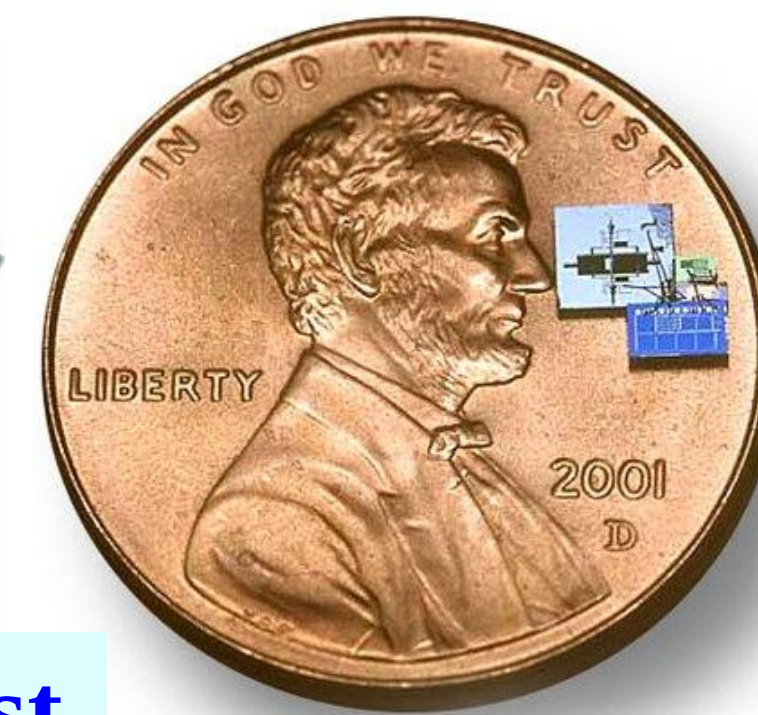
□ Short Transmission Range

- 30–100 meters

Tmote Sky



Smart Dust



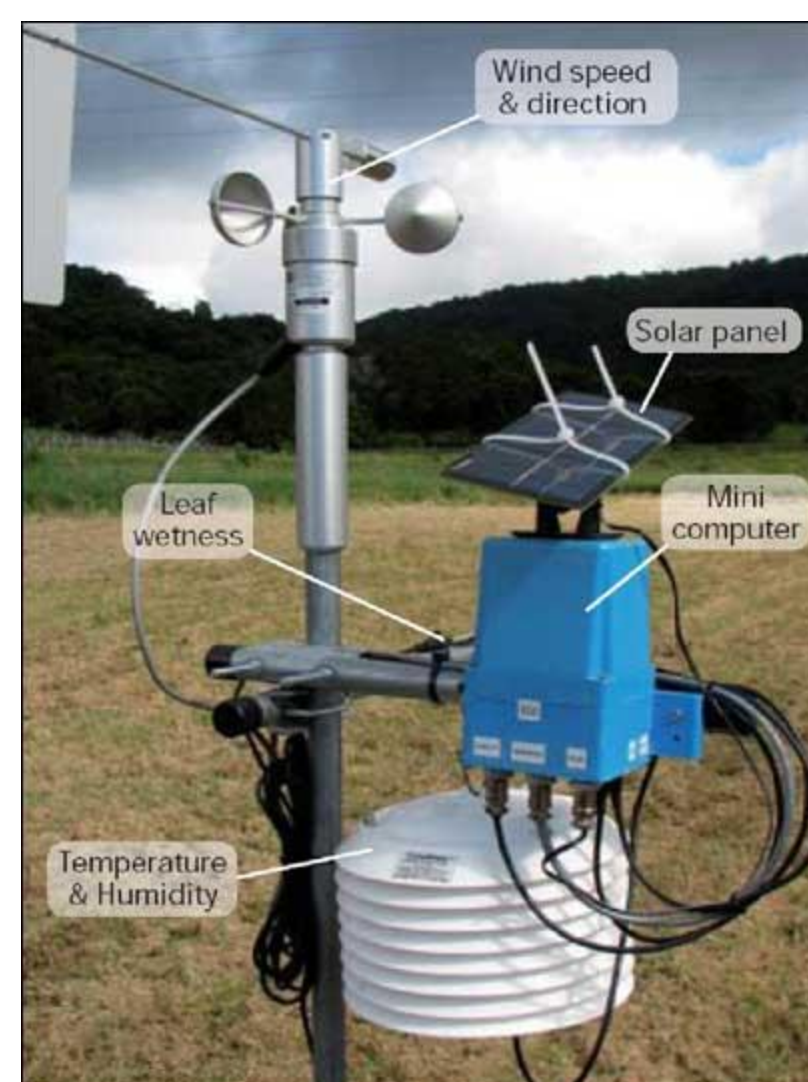
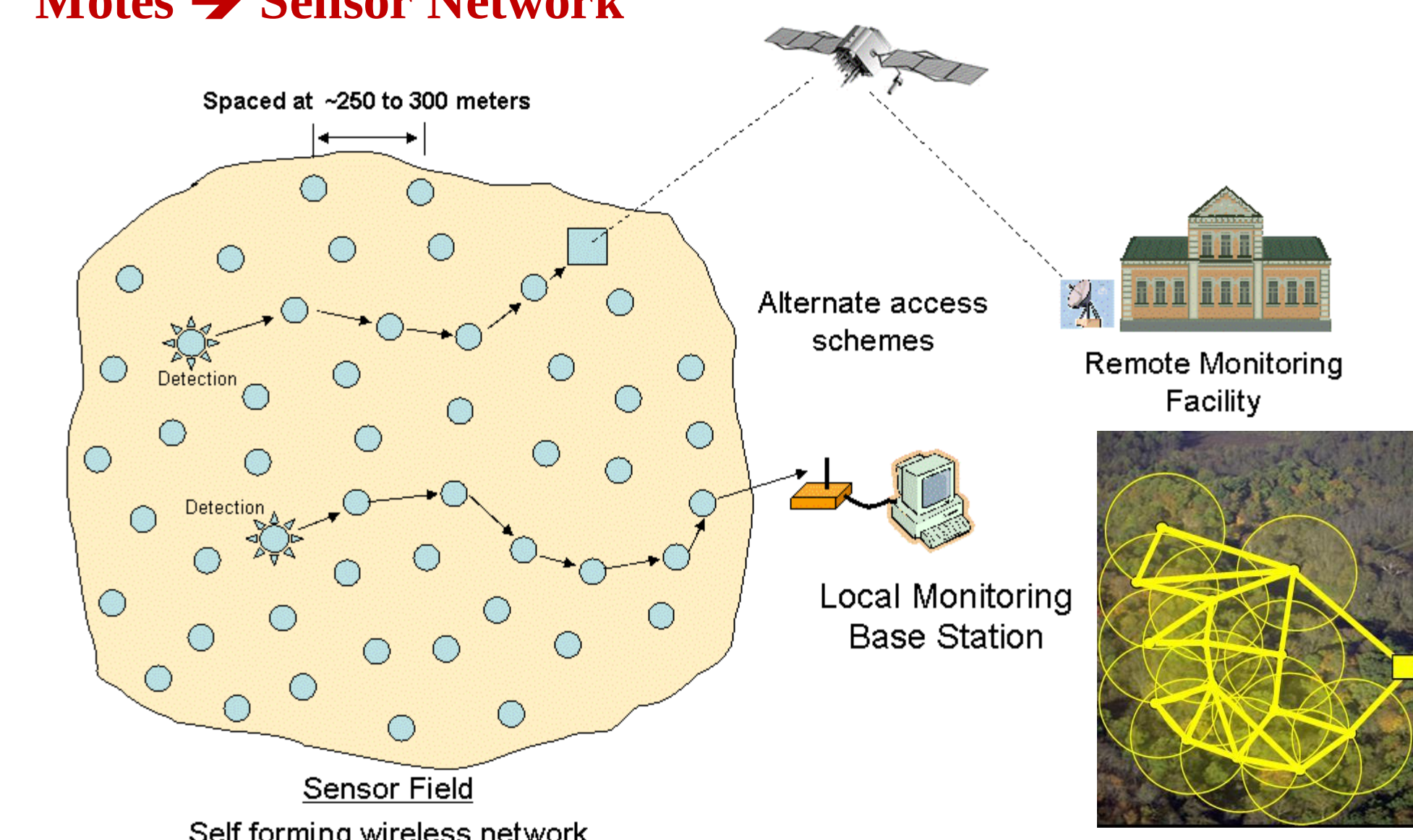
MICA2 Dot



Sun SPOT

What can we do with MOTES?

Form a Network with these Motes → Sensor Network



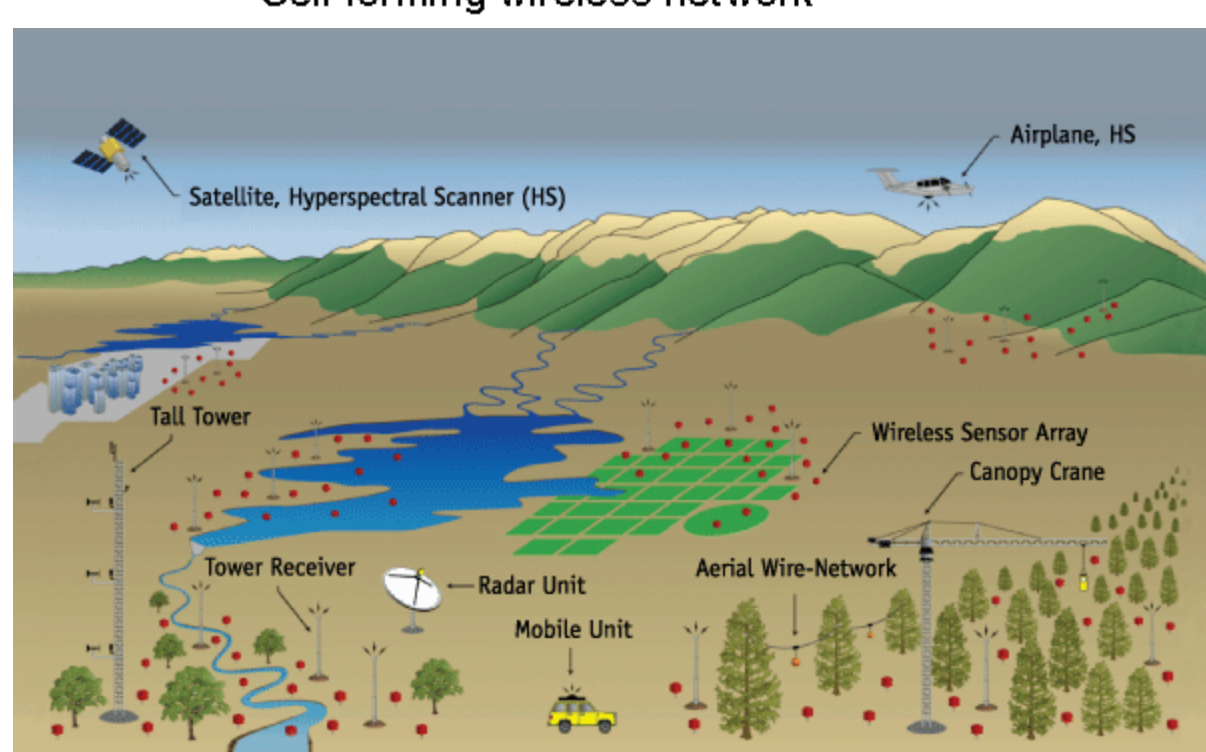
Precision Agriculture – Monitor temperature, light, soil quality, humidity



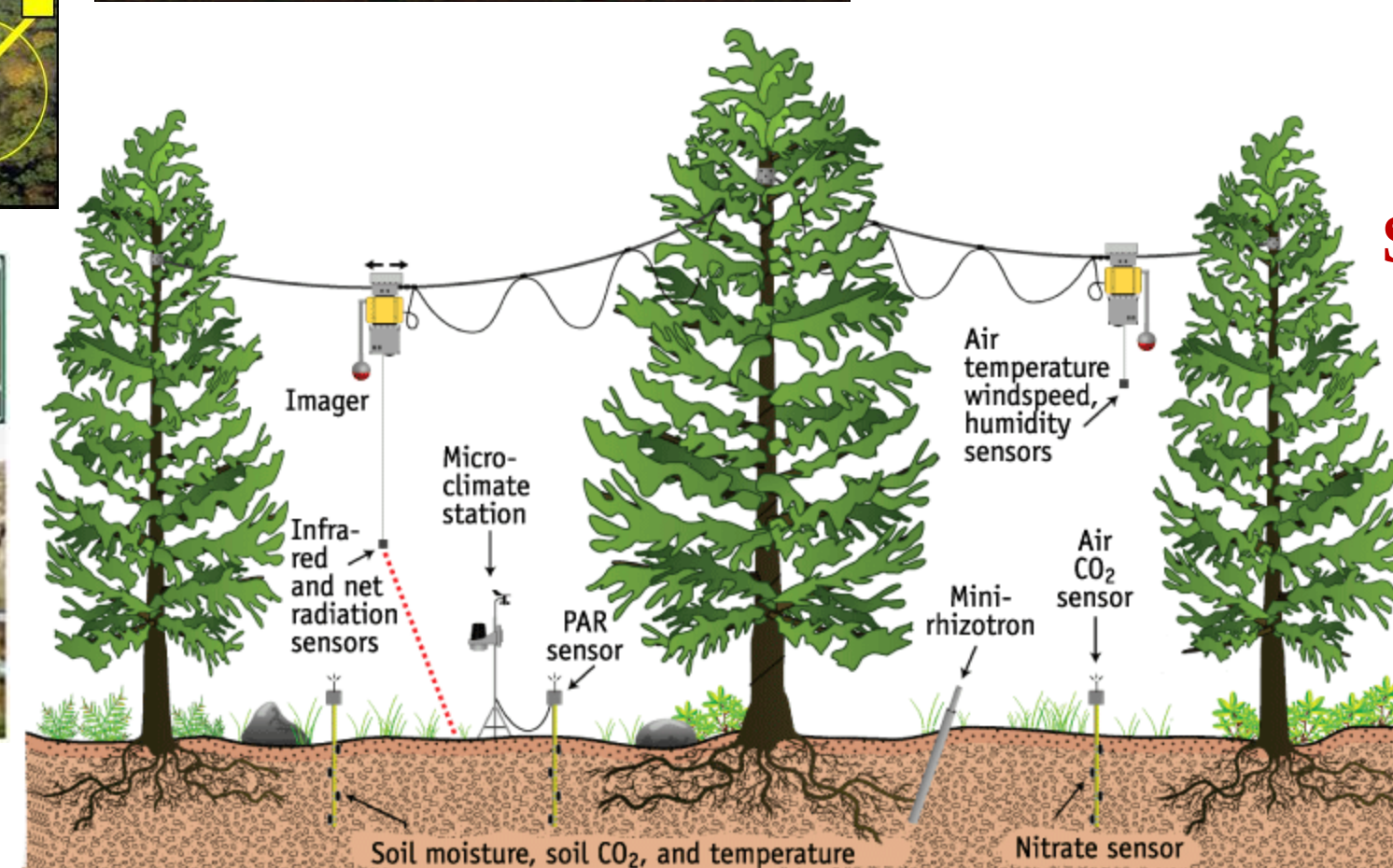
“Talking” Plants! – Send text messages when thirsty



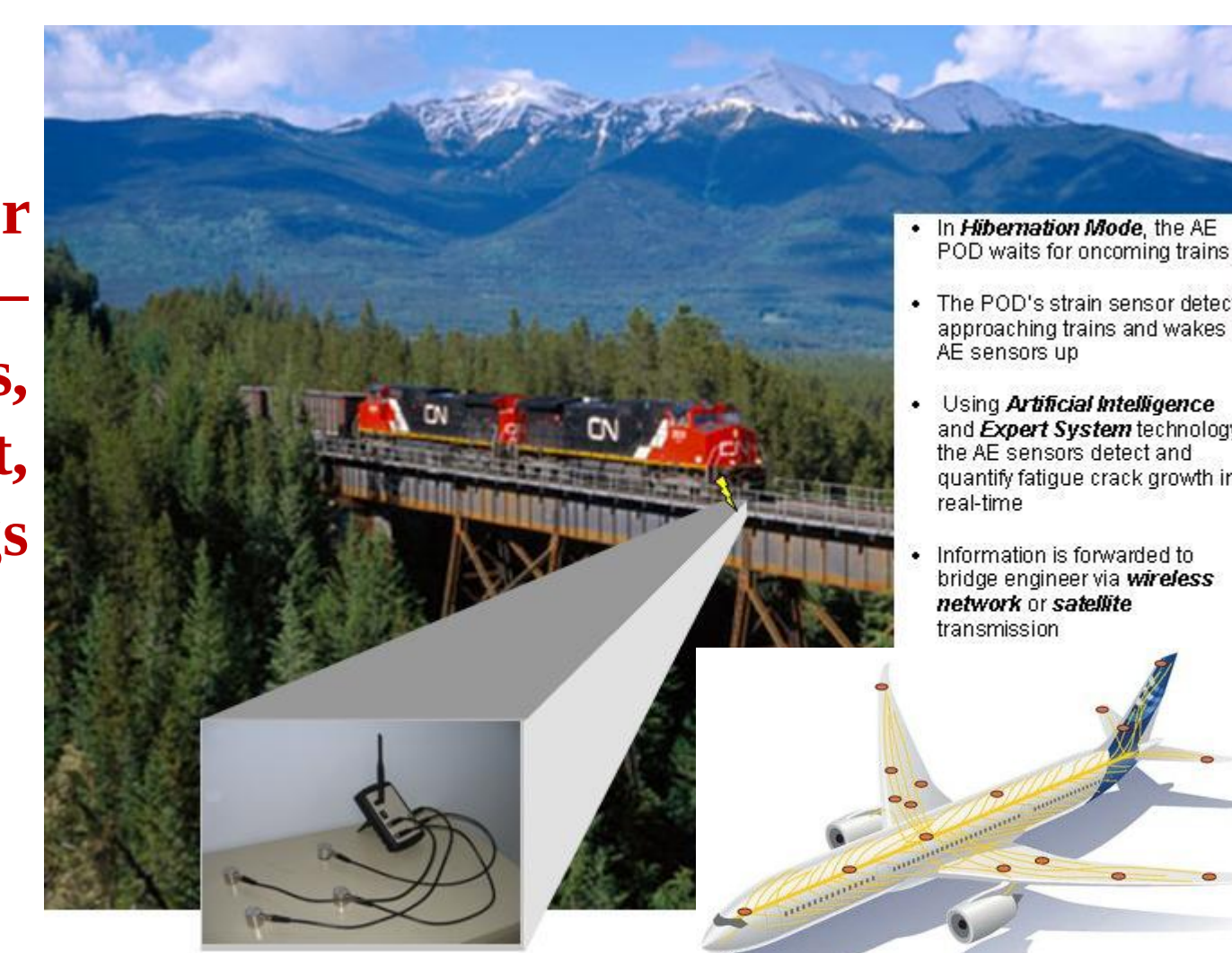
Wildfire Monitoring - Early warnings to save lives and property



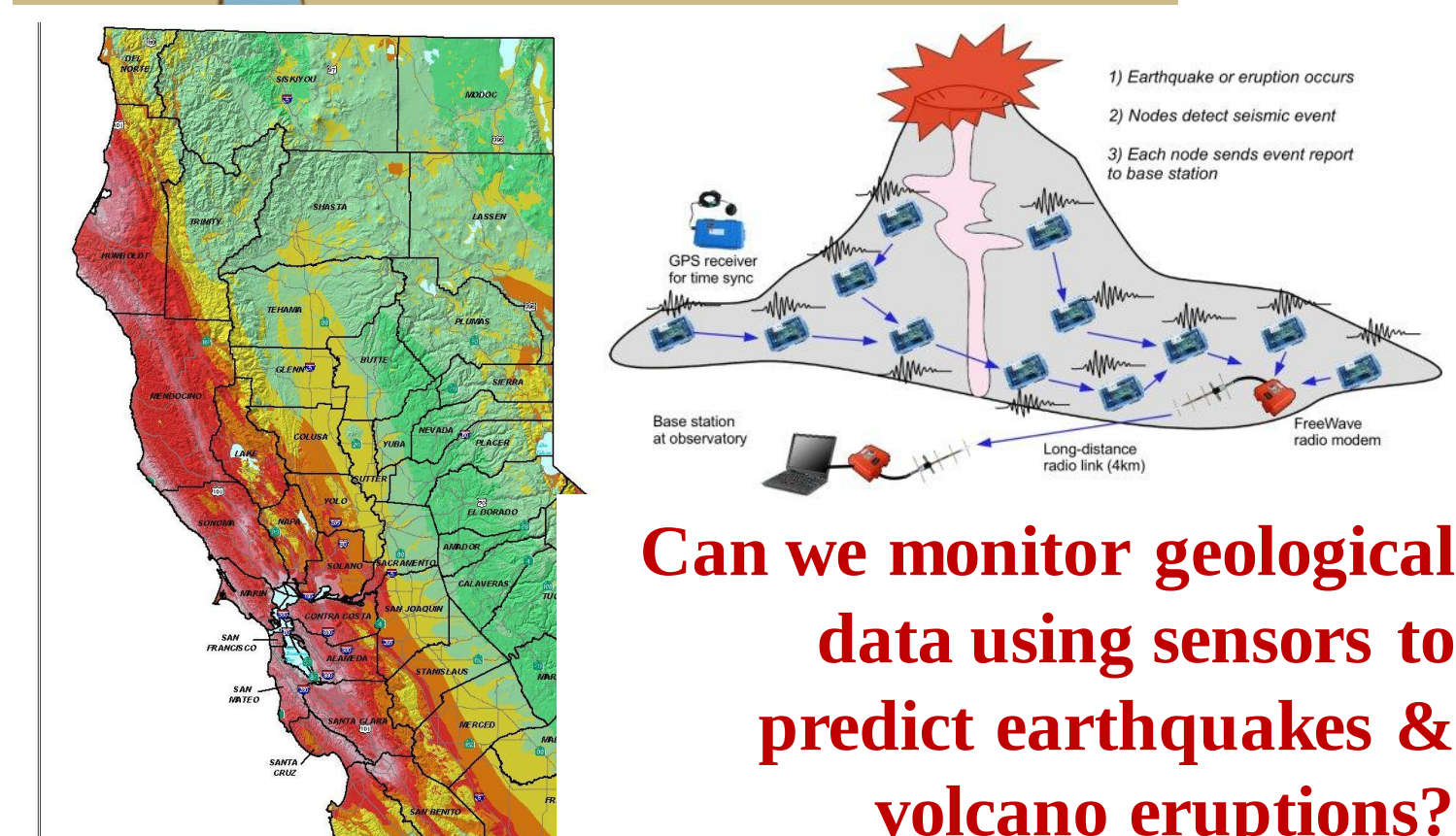
Military Networks



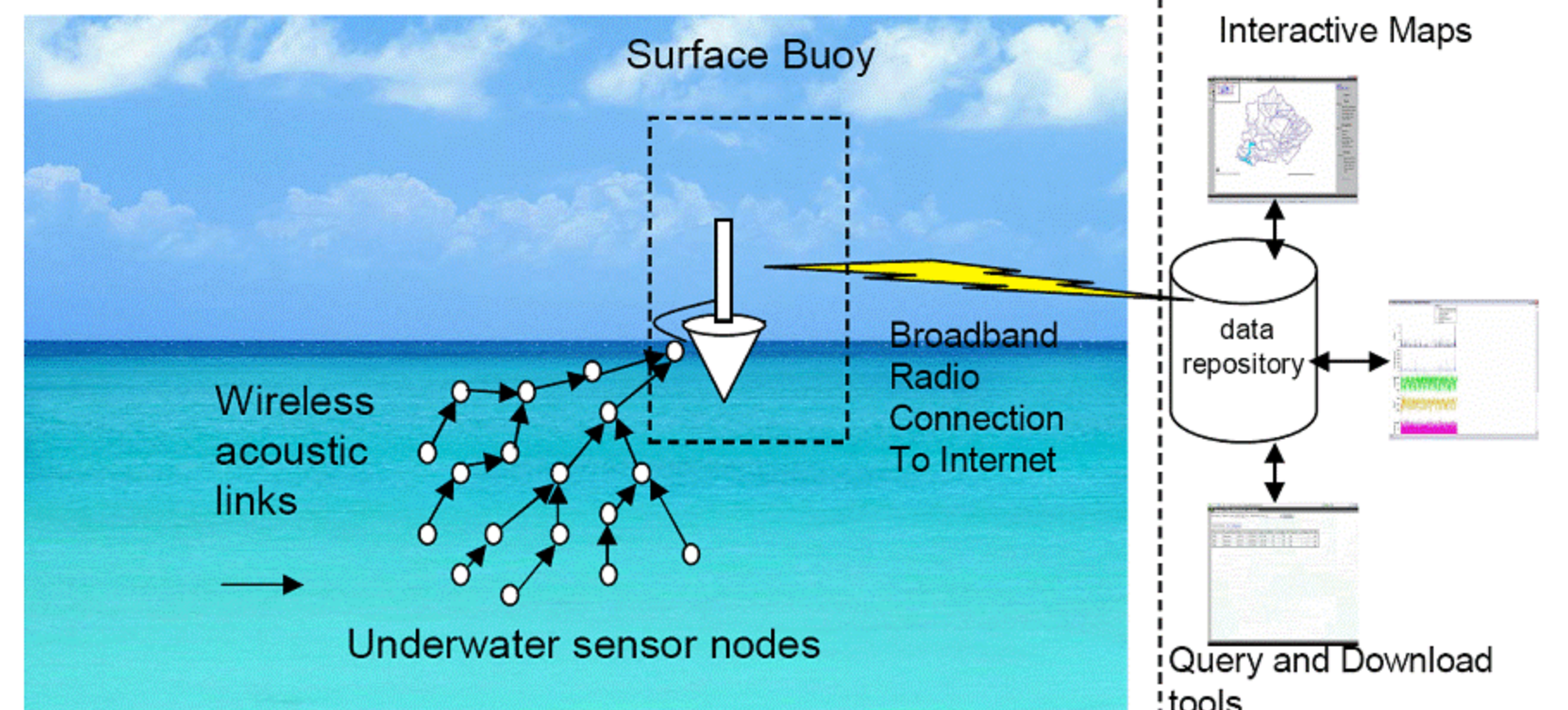
Monitor Structures – bridges, aircraft, buildings



- In Information Mode, the AE POD waits for oncoming trains
- The POD's strain sensor detects approaching trains and wakes AE sensors up
- Using Artificial Intelligence and Expert Systems technology, the AE sensors detect and quantify fatigue crack growth in real-time
- Information is forwarded to bridge engineer via wireless network or satellite transmission

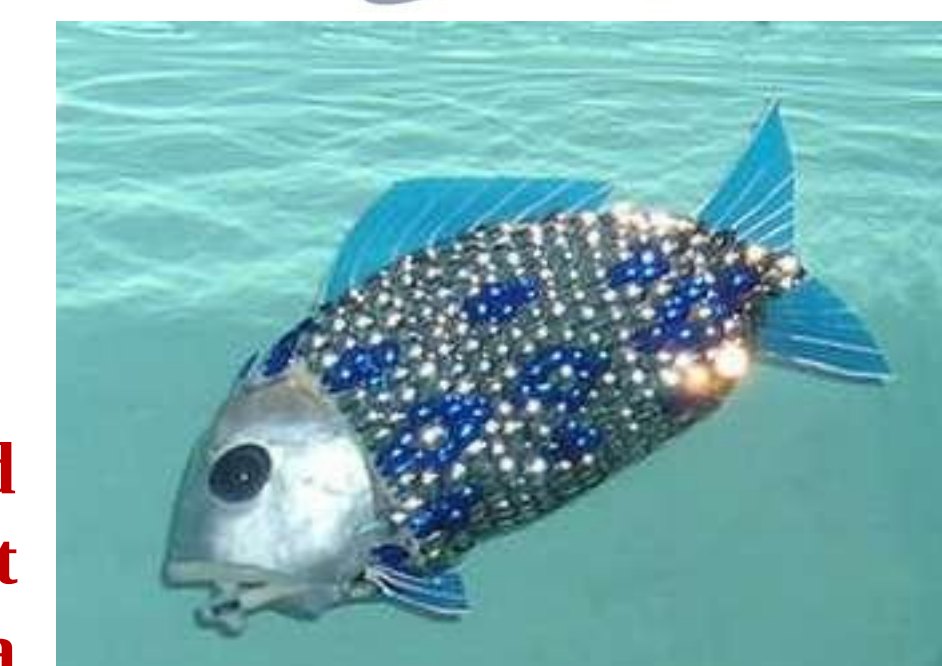


Can we monitor geological data using sensors to predict earthquakes & volcano eruptions?



Underwater Sensor Networks - monitor marine life (e.g. coral reef, fishes)

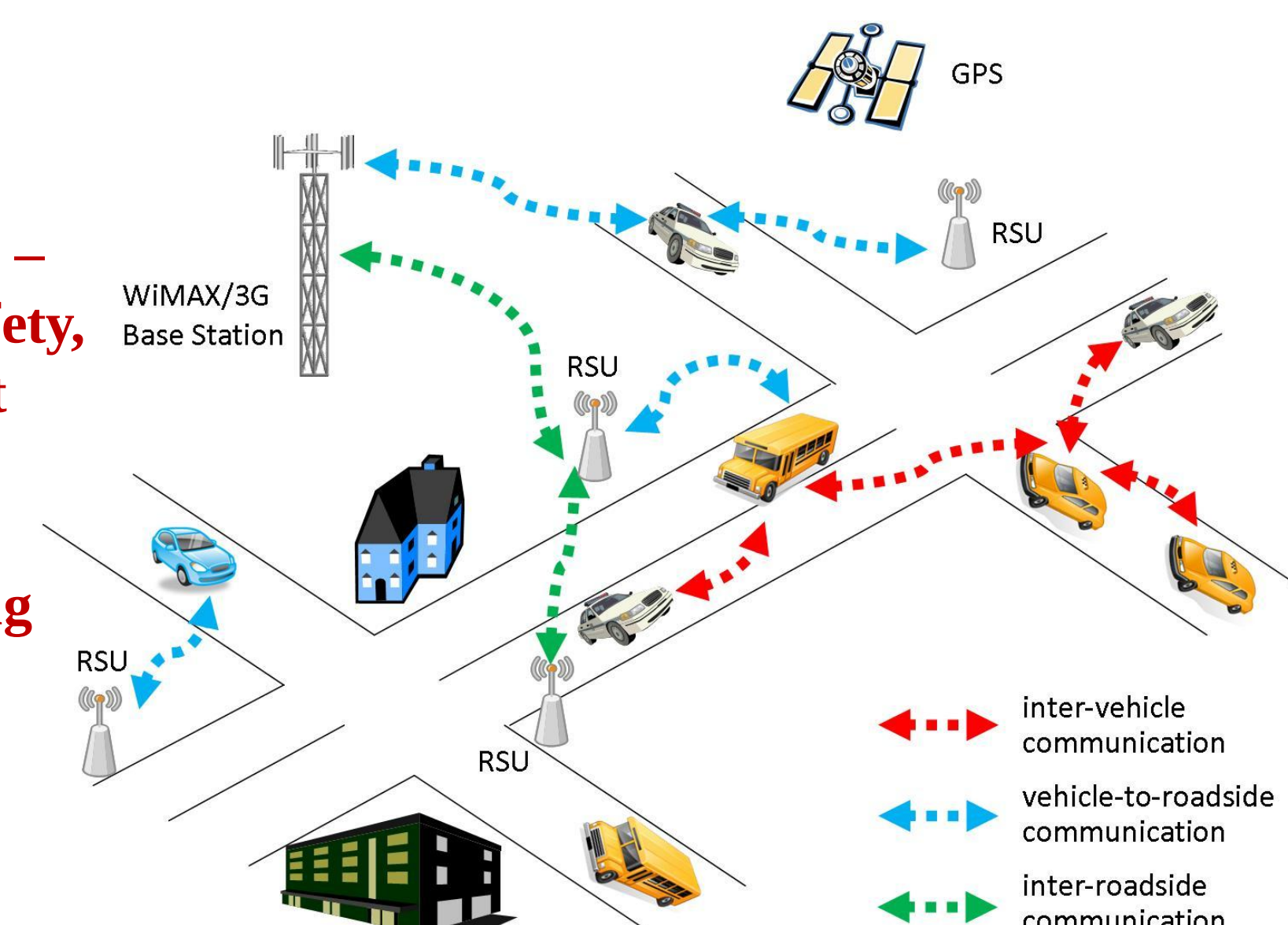
Robot fish equipped with sensors to collect data



Body Sensor Networks – monitor blood pressure, heart, temperature



Vehicular Networks – traffic safety, intelligent parking, remote monitoring of homes



- <http://anrg.usc.edu>
- <http://research.cens.ucla.edu>
- <http://www.xbow.com>
- <http://enl.usc.edu>
- <http://vehicularlab.org>